

Chapter	Title	Total questions	Type
Ch 8	Journey Inside the Atom	12	Formative

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student “understood” a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by sections, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

SECTION 1

8.1 Rediscovering the Roots of Atomic Theory

Q1 · Learning Outcome

Students can distinguish between philosophical reasoning and evidence-based scientific argument by identifying the specific evidential criteria that one framework provides and the other lacks.

Cognitive Demand

Understanding

Acharya Kanada and the Greek philosopher Democritus both proposed that matter consists of tiny, indivisible particles. Dalton, in 1808, also proposed indivisible atoms. What does Dalton's proposal have that the ancient accounts lack?

- A. Dalton gave atoms specific names derived from Sanskrit and Greek roots.
- B. Dalton was the first to propose that matter is made of indivisible particles, while Kanada and Democritus did not.
- C. **Dalton's atoms are based on experimental evidence and specific claims about how atoms of different elements combine in fixed ratios. ✓**
- D. Dalton's atoms are spherical in shape, whereas ancient philosophers described atoms as having different geometrical forms.

Answer section

Correct answer: C

What each choice reveals

- A. Conflates naming conventions with the nature of the evidential basis — the distinction the question tests is philosophical vs experimental, not etymological origin.
- B. Directly contradicts the chapter content, which states both ancient traditions proposed indivisible particles independently before Dalton — selecting this reveals a factual misread of the section.
- D. Introduces a shape detail not present in section 8.1 and misidentifies what changed between ancient and Daltonian atomism — the change is in evidential grounding, not geometry.

SECTION 2

8.2 A Short Historical Journey Through Atomic Models

Q2 · Learning Outcome

Students can evaluate a scientific model by identifying the evidence that supports it and the questions it leaves unresolved, using an analogy to expose the model's structural assumptions.

Cognitive Demand

Analysis

Thomson's cathode ray experiment provided two key pieces of evidence that led him to propose the plum pudding model of the atom.

- (a) State the two pieces of evidence and the inference each one supports.

- (b) The plum pudding model is often compared to a watermelon — red pulp as positive matter, seeds as electrons. Identify one aspect of atomic structure the watermelon analogy represents correctly and one structural question about the atom the analogy cannot answer.

Answer section

Look for

- Correctly identifies both pieces of evidence: cathode rays deflect toward the positive plate (negative charge) AND cathode rays are the same regardless of cathode material (electrons are universal).
- Links each piece of evidence to its specific inference rather than restating it: deflection → electrons are negatively charged; independence from material → electrons are present in all elements.
- Correctly identifies the watermelon analogy's strength: it represents the coexistence of positive and negative components in a single neutral structure.
- Correctly identifies the analogy's limitation: it cannot explain where the positive charge comes from, what holds the electrons in specific positions, or how the model would be tested experimentally — any one of these counts.
- Reasoning is explicit: the student states why each inference follows from the evidence, not just what the inference is.

SECTION 3

8.2.2 Testing Thomson's model: The gold foil experiment

Q3 · Learning Outcome

Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections.

Cognitive Demand

Analysis

In the gold foil experiment, most alpha particles passed straight through the foil, a small number deflected at large angles, and a very few bounced directly back. Which observation is sufficient on its own to falsify Thomson's plum pudding model, and what structural feature of the atom does it imply?

- A. A few particles bouncing directly back — this implies a tiny, dense, positively charged nucleus, which Thomson's diffuse positive sphere cannot produce. ✓**
- B.** Most particles passing straight through — this shows the atom is mostly empty space, which Thomson's model did not predict.
- C.** Some particles deflecting at large angles — this shows electrons are negatively charged, confirming Thomson's placement of negative charges inside the atom.
- D.** The combination of all three observations together is required; no single observation alone is sufficient to falsify Thomson's model.

Answer section

Correct answer: A

What each choice reveals

- B** Confuses the inference from undeflected particles (empty space) with what falsifies Thomson's model — Thomson's model did predict slight deflections from diffuse charge, so undeflected particles alone are not sufficient to falsify it; the backscattering is what no diffuse model can explain.
- C** Misidentifies large-angle deflection as confirming Thomson's electron placement rather than challenging Thomson's positive-charge distribution; also incorrectly attributes large-angle deflection to electron behaviour rather than nuclear concentration.
- D** Misunderstands the logic of falsification — in science, one sufficiently anomalous observation falsifies a model; demanding all three together reflects a misunderstanding of how experimental evidence works against theoretical predictions.

Q4 · Learning Outcome

Students can construct a logical argument showing how a single anomalous experimental result falsifies a model, and derive structural inferences about a system from the pattern of observed deflections.

Cognitive Demand

Application

Rutherford's planetary model drew three conclusions from the gold foil experiment. However, a physics principle requires a charged particle moving in a circular orbit to continuously radiate energy.

- (a) State the three conclusions Rutherford drew.

- (b) Use the physics principle stated above to explain why an electron orbiting the nucleus according to Rutherford's model would be unstable.
- (c) A helium atom has 2 protons. State the number of electrons in a neutral helium atom and explain why.

Answer section

Expected elements

- Three conclusions: most of the atom is empty space; all positive charge and most mass are concentrated in a tiny nucleus; electrons revolve around the nucleus in orbits.
- Instability argument: because the orbiting electron is accelerating (circular motion), it must radiate energy; losing energy causes it to spiral inward and collapse into the nucleus.
- Helium has 2 electrons in the neutral atom; the number of electrons equals the number of protons because the atom is electrically neutral (charges must balance).

SECTION 4

8.2.3 Bohr's model of the atom

Q5 · Learning Outcome

Students can explain how introducing the concept of fixed energy levels resolves a contradiction in a prior model by identifying the specific physical consequence the new postulate prevents.

Cognitive Demand

Understanding

Bohr proposed that electrons occupy fixed circular paths called stationary states, each with a definite constant energy. Which statement correctly explains how this postulate resolves the problem with Rutherford's model?

- A. Bohr's shells are labelled K, L, M, N because earlier letters were reserved for a possible inner series, which prevents energy loss from the atom.
- B. Electrons in Bohr's model can only move between shells when they absorb energy, so they naturally remain in the lowest shell and never spiral inward.
- C. **In a stationary state, the electron moves but does not radiate energy, so it maintains its orbit indefinitely without spiralling into the nucleus. ✓**
- D. In a stationary state, the electron stops moving, so it no longer accelerates and does not radiate energy.

Answer section

Correct answer: C

What each choice reveals

- D Conflates 'stationary state' with the electron being literally stationary — a common and important misconception; students selecting this have not distinguished between the electron's motion and its energy state.
- A Confuses the historical origin of the K-N labelling convention (an interesting but structurally unrelated fact) with the physical mechanism that prevents energy loss — a confusion between a naming convention and a physical postulate.
- B Partially correct (electrons do need to absorb energy to move to a higher shell) but explains why electrons stay in one shell by appeal to the absorption rule rather than the stationary-state postulate; it also implies an electron cannot be in any shell above K by default, which is wrong for multi-electron atoms.

SECTION 5

8.3 What Components Contribute to the Mass of an Atom?

Q6 · Learning Outcome

Students can resolve a quantitative discrepancy in a scientific model by identifying the missing component and explaining the physical role that component plays in the system's stability.

Cognitive Demand

Application

Hydrogen has 1 proton and helium has 2 protons, yet helium is approximately four times as heavy as hydrogen, not twice as heavy.

- (a) Identify the particle discovered by Chadwick that accounts for this discrepancy and state its charge and approximate mass.

(b) Uranium has 92 protons and 146 neutrons. Explain, using the information about neutrons from this chapter, why uranium needs far more neutrons than protons to remain stable.

Answer section

Expected elements

- Neutron: charge 0 (neutral), mass approximately equal to a proton (~1u); found in the nucleus.
- Helium has 2 protons and 2 neutrons, giving mass ≈ 4u, resolving the discrepancy — mass comes from both protons and neutrons.
- Protons repel each other electrostatically; neutrons have no charge so they dilute the repulsion by spacing protons apart and strengthen the nuclear binding force — heavier nuclei need progressively more neutrons to counterbalance the increased proton repulsion.

SECTION 6
8.4 Symbols of Elements

Q7 · Learning Outcome Students can apply a rule-governed symbolic notation system to encode and decode quantitative information about atomic composition from a standardised representation.	Cognitive Demand Application
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The nuclide notation for an atom is written as ²³₁₁Na. Using this notation, how many neutrons does this atom contain?

- A. 11
- B. 12 ✓
- C. 23
- D. 34

Answer section

Correct answer: B

What each choice reveals

- A Reads the subscript (Z=11) as the neutron count, confusing atomic number with neutron number — the student has not applied A-Z.
- C Reads the superscript (A=23) as the neutron count, confusing mass number with neutron number — the student has not subtracted Z from A.
- D Adds A and Z (23+11=34) instead of subtracting — a procedural error revealing the student knows both numbers are relevant but has applied the wrong operation.

SECTION 7
8.6 Mass Number

Q8 · Learning Outcome Students can apply a multi-variable numerical relationship to determine unknown subatomic quantities, showing each step of the reasoning chain explicitly.	Cognitive Demand Application
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Refer to the table provided. Complete all missing values for each atom, showing the relationship you use for each row.

Atom	Atomic Number (Z)	Mass Number (A)	Protons	Neutrons	Electrons
Carbon	6	12	?	?	?
Sulfur	16	32	?	?	?
Chlorine	17	?	17	18	?
Calcium	?	40	20	?	20

Answer section

Expected elements

- Carbon: protons = 6 (equals Z), neutrons = A-Z = 12-6 = 6, electrons = 6 (equals Z for neutral atom).

- Sulfur: protons = 16, neutrons = $32-16 = 16$, electrons = 16.
- Chlorine: $A = \text{protons} + \text{neutrons} = 17+18 = 35$; electrons = 17 (equals Z).
- Calcium: $Z = 20$ (equals proton count given); neutrons = $A-Z = 40-20 = 20$.
- Student explicitly states the relationship used for each calculation ($A-Z$ for neutrons; Z equals electrons for neutral atom; $A = \text{protons} + \text{neutrons}$ for mass number).

SECTION 8

8.7 How Are Electrons Distributed in Different Energy Levels?

Q9 · Learning Outcome

Students can construct the electron shell distribution of an atom by systematically applying capacity and filling-order rules, and identify the valence shell from the resulting configuration.

Cognitive Demand

Application

An atom has atomic number 16.

- (a) Write the electronic configuration of this atom, showing the number of electrons in each shell.
- (b) Identify the valence shell and state the number of valence electrons.
- (c) A student writes the configuration of the atom with atomic number 18 as 2, 8, 8. Another student writes it as 2, 16. State which is correct and explain the rule that makes the second configuration impossible.

Answer section

Expected elements

- Atomic number 16: $K=2$, $L=8$, $M=6$. Configuration 2, 8, 6.
- Valence shell is M; 6 valence electrons.
- Configuration 2, 8, 8 is correct for $Z=18$; the second (2, 16) is impossible because the outermost shell cannot hold more than 8 electrons regardless of its $2n^2$ capacity.

SECTION 9

8.8 Combining Capacity of an Atom: Valency

Q10 · Learning Outcome

Students can predict the combining capacity of an atom by applying the octet rule to its electronic configuration and distinguishing between electron loss, gain, and sharing pathways.

Cognitive Demand

Application

Magnesium has atomic number 12. Its electronic configuration is 2, 8, 2. What is the valency of magnesium and what does the atom do to achieve a stable configuration?

- A. Valency 2; magnesium gains 2 electrons, filling its M shell to 8.
- B. Valency 2; magnesium loses 2 electrons, leaving a complete L shell as the new outermost shell. ✓**
- C. Valency 6; magnesium gains 6 electrons to fill its outermost shell to 8.
- D. Valency 8; magnesium shares 8 electrons to reach an octet.

Answer section

Correct answer: B

What each choice reveals

- C Treats the 2 valence electrons as deficient by 6 (as if $8-2=6$ electrons must be added), confusing gaining with the number needed to fill — also incorrectly applies gaining rather than losing for an atom with only 2 valence electrons.
- A Correctly identifies the valency as 2 but incorrectly assigns the action as gaining rather than losing; a student selecting this knows valency = 2 but has confused loss and gain for this configuration.
- D Applies the number 8 (the target octet) as the valency rather than deriving valency as the number of electrons actually transferred or shared — a confusion between the target configuration and the action needed to reach it.

SECTION 10

8.9 A Deeper Look into Atomic Structure

Q11 · Learning Outcome

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

Cognitive Demand

Application

Refer to the table provided.

- (a) Calculate the weighted average atomic mass of boron. Show your working.
- (b) Does any single boron atom in nature have the mass you calculated? Explain your answer.
- (c) Boron-10 and Boron-11 are isotopes of the same element. State two quantities that are the same for both atoms and one quantity that differs.

Isotope	Mass Number	Natural Abundance (%)
Boron-10	10	19.9
Boron-11	11	80.1

Answer section**Expected elements**

- Weighted average = $(10 \times 19.9/100) + (11 \times 80.1/100) = 1.99 + 8.811 = 10.8 \text{ u}$ (accept values rounded correctly to one decimal place).
- No single boron atom has mass 10.8 u; the value is a weighted average representing the isotopic mixture across a large population of boron atoms, not the mass of any individual atom.
- Same: atomic number ($Z=5$), number of electrons (5), chemical properties. Different: mass number (A), number of neutrons (5 vs 6). (Any two same and one different accepted.)

Q12 · Learning Outcome

Students can classify atomic variants by their conserved and differing quantities, and calculate a statistically weighted average to represent a naturally occurring mixture of isotopic masses.

Cognitive Demand

Analysis

The chapter states that isotopes of an element have the same chemical properties but different physical properties. Argon ($Z=18$, $A=40$), potassium ($Z=19$, $A=40$), and calcium ($Z=20$, $A=40$) are described as isobars.

- (a) Explain, using the concept of electronic configuration, why the two isotopes of carbon ($^{12}_6\text{C}$ and $^{13}_6\text{C}$) have identical chemical properties despite having different masses.
- (b) Argon, potassium, and calcium are called isobars, not isotopes. Construct an argument — using specific values of Z and A — distinguishing why these three atoms cannot be isotopes of the same element, and what structural feature they share that makes them isobars.

Answer section**Look for**

- Correctly identifies that both carbon isotopes have $Z=6$ and therefore the same electronic configuration (2,4), which determines chemical behaviour — chemical properties depend on electron arrangement, not nuclear mass.
- Correctly distinguishes: isotopes require the same Z ; argon ($Z=18$), potassium ($Z=19$), and calcium ($Z=20$) all have different atomic numbers, so they cannot be isotopes.
- Correctly identifies the isobar feature: all three have $A=40$ (same total number of nucleons) but different atomic numbers — same mass number, different identities.
- Argument uses specific numerical values of Z and A , not just the definitions.
- Reasoning is logically sequenced: the student moves from the definition to the specific values to the conclusion, not from the conclusion backward.